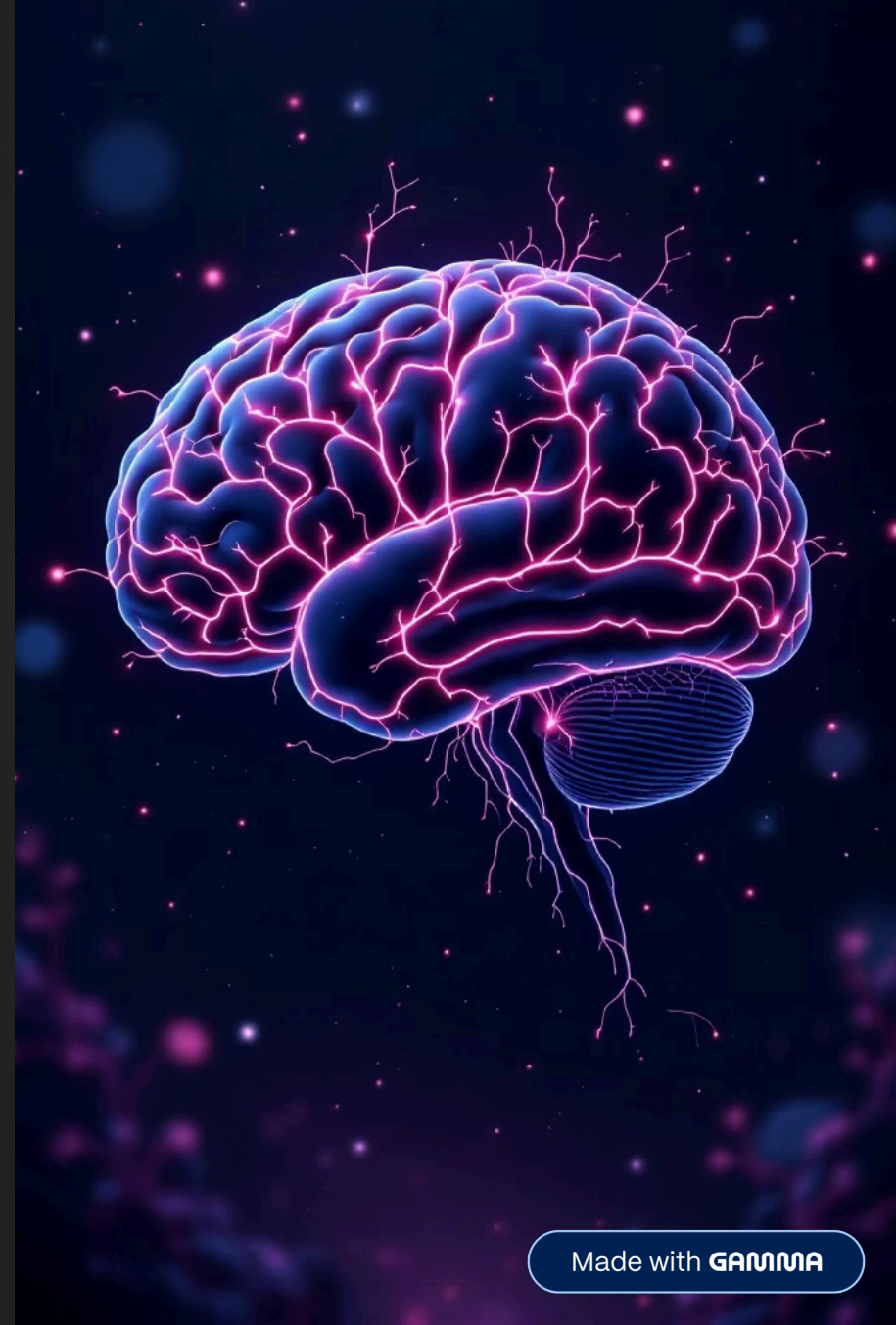


The Human Brain: A Universe Within

An exploration of the most complex structure in the known universe



The Brain's Insatiable Appetite

2%

Body Weight

The brain occupies just 2% of total body mass

20-25%

Energy Consumption

Consumes 20-25% of the body's glucose and oxygen

10s

Energy Cut-Off

Just 10 seconds without energy causes complete functional cessation

This remarkable organ demands constant fuel to maintain consciousness, thought, and bodily control.

A Jell-O-Like Marvel, Fiercely Protected



Natural Defence System

Despite its delicate, jello-like consistency (attributed to high fat content), the brain is shielded by multiple protective layers working in concert.

- Bony skull provides physical barrier
- Meninges create protective membranes
- Cerebrospinal fluid cushions against impact
- Blood-Brain Barrier acts as selective gatekeeper

Debunking the Myths: What We *Don't* Know

The Triune Brain Theory

Once thought to have evolved in three distinct layers, modern neuroscience reveals the brain operates as an integrated whole, not separate evolutionary stages.

The 10% Myth

We use virtually every part of our brain. Even during simple tasks, widespread neural activity shows no dormant regions waiting to be unlocked.

Brain Plasticity

Contrary to old beliefs, the adult brain remains remarkably adaptable throughout life, capable of forming new connections and reorganising itself.

The Cerebral Cortex: Our Seat of Consciousness

Frontal Lobe

Speech production, executive thought, decision-making, and emotional regulation

Parietal Lobe

Processing sensory information and spatial awareness

Occipital Lobe

Visual processing and interpretation of what we see

Temporal Lobe

Hearing, language comprehension, and memory formation

Comprising 80% of the brain's mass but only 19% of its neurons, this wrinkled outer layer enables our most sophisticated cognitive abilities.



The Primate Advantage: More Neurons, More Power

Density Matters More Than Size

Whilst other animals may possess larger brains, primates—especially humans—pack significantly more neurons into their cerebral cortex per unit volume.

This primate advantage allows for greater cognitive complexity, enhanced processing power, and the sophisticated problem-solving abilities that distinguish human thought.

The secret lies not in size, but in neural density and connectivity.



The Astonishing Scale of Our Neural Network

86B

Neurons

Approximately 86 billion neurons form the brain's communication network

85B

Glial Cells

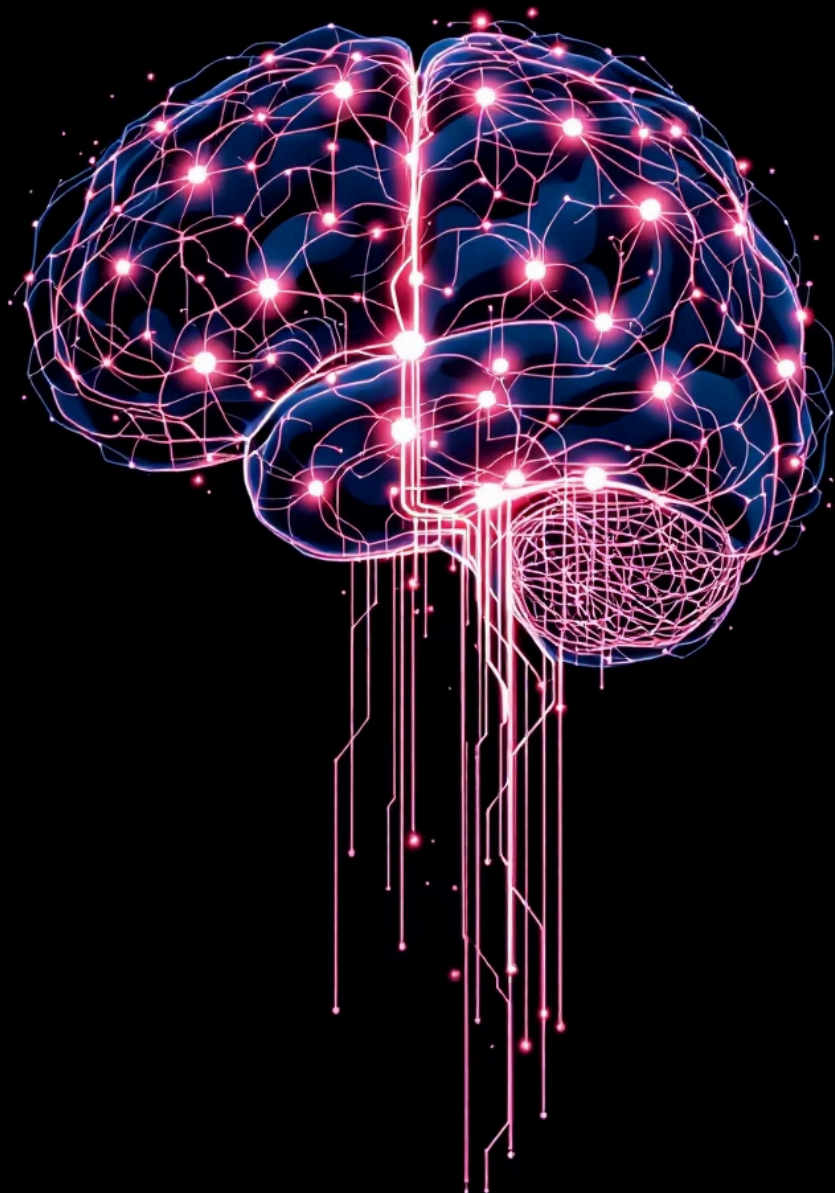
Roughly 85 billion glial cells support and maintain neural function

100T

Connections

Estimated 100 trillion synaptic connections enable complex thought

For every neuron, there are roughly an equal number of glial cells—supporting cells that nourish, protect, and maintain the neural network.



A Symphony of Lobes: Specialisation and Integration



Left Hemisphere

Logic and analytical thinking, language processing, step-by-step reasoning, and detailed sequential processing



Dynamic Interplay

Both hemispheres communicate constantly through the corpus callosum, integrating specialised functions



Right Hemisphere

Holistic thought and intuition, creativity and artistic ability, big-picture understanding and spatial awareness

Key Structures and Their Roles



Cerebrum

The largest part of the brain, divided into left and right hemispheres, each containing four distinct lobes responsible for higher cognitive functions, sensory processing, and voluntary movement.



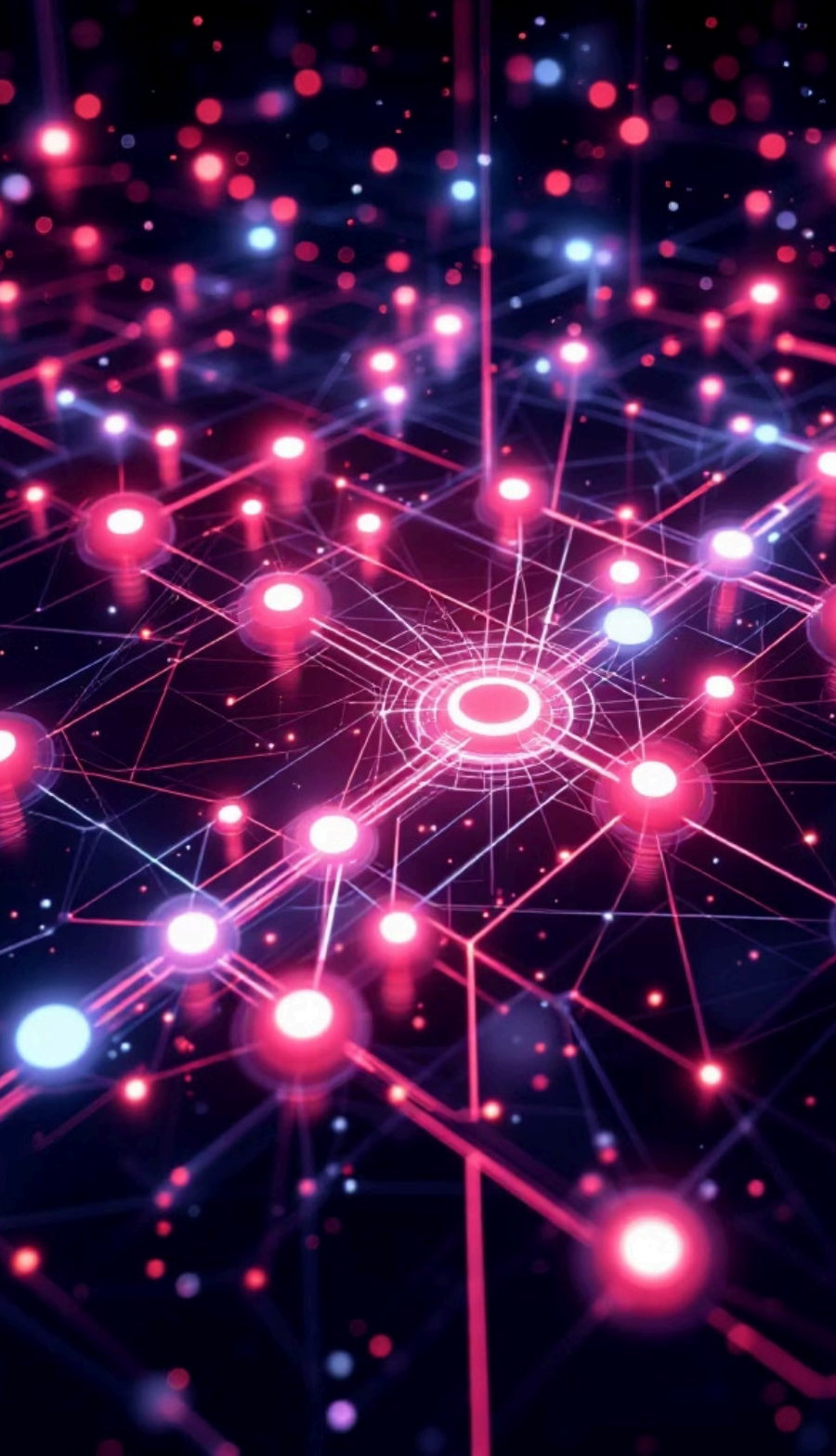
Cerebellum

Located at the rear base of the brain, this structure is crucial for motor coordination, balance, and fine-tuning movements. It ensures smooth, precise physical actions.



Brainstem

Connecting the cerebrum and cerebellum to the spinal cord, this vital structure controls automatic functions essential for survival, including breathing, heart rate, and sleep cycles.



The Brain: Our Ultimate Frontier



Energy-Hungry

Demands constant fuel
despite small size



Resilient

Protected yet remarkably
adaptable



Mysterious

Continues to reveal new discoveries

A complex, energy-hungry, yet incredibly resilient organ that remains a source of wonder and scientific discovery, constantly pushing the boundaries of our understanding.